

Joel Hempel

Machine Learning and Full-Stack Development

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Education

BSc, Computer Science

University of British Columbia

Vancouver, Canada

2022–2026

- **GPA:** 4.00/4.33

Transfer Credits

Langara College

Vancouver, Canada

2020–2022

- **GPA:** 4.17/4.33
- **Awards:**
 - Ernest E. Livesey Memorial Prize in Mathematics
 - LSU Faculty of Science Award (highest cumulative GPA)

Technologies

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS, Erlang

Frameworks & Libraries: PyTorch, React, Next.js, FastAPI, CUDA

Web/Databases: Supabase, SQL, Tailwind, PHP

Tools: Git, Docker, AWS, Google Cloud, JUnit, Google Colab

ML: Traditional Models, Neural Networks, Computer Vision, LLMs

Experience

SWE Intern (ML & Full-Stack)

Metrized Consulting Inc.

Vancouver, BC

09/2023 – 08/2024

- **Built and deployed an alpha version of a computer vision powered cost-estimation web-application to reduce contractor estimation time by around 80%.**
- Co-created a distributed back-end solution for separate training, inference, and core servers—allowing for easier scaling.
- Engineered a ground-truth dataset of **10,000 images**, reduced synthetic data generation/augmentation time by **95%**, and trained a **top-performing computer vision model with 95% test-set accuracy**.
- Authored comprehensive documentation (on **20,000+** lines of code across 4 repositories), streamlining new-hire onboarding.
- Expanded an existing **front-end by adding new 10 new features** and improving user-experience.
- Proposed a **breakthrough feature** for AI-assisted symbol labeling **increasing accuracy by 400%**.
- Conducted experimental research on language models for potential future products.
- **Technologies:** Python, PyTorch, Next.js, AWS, Supabase, JavaScript, YOLO, SAM, LLMs

Undergraduate Research Assistant

Systopia Lab

Vancouver, BC

05/2023 – 12/2023

- Researched spring-onset detection using satellite data and flower proxies combined with machine learning.
- Developed a data pre-processing pipeline to transform temperature data and built the highest-performing model for the paper **reducing the in-region error of the state-of-the-art first-bloom index by 16.9%**.
- Leveraged the resulting data for deep-learning models and transfer-learning experiments.
- **Co-authored and [presented a paper at NeurIPS 2023](#)**
- **Technologies:** Python, scikit-learn

Software Sub-Team

UBC Subbots

Vancouver, BC

09/2024 – Present

- Building an autonomous underwater vehicle (AUV) for the RoboSub 2025 competition.
- Implementing trajectory-generation algorithms (ROS2, Gazebo) to guide the AUV around buoys.
- Improving IMU reliability by evaluating hardware options and refining data extraction for real-time sensor fusion.
- **Technologies:** C++, ROS2, Gazebo, Arduino